SYSTEM LOGGING AND REPORTING

Operating System

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SYSTEM LOGGING AND REPORTING FRAMEWORK FOR LINUX-BASED SYSTEMS

Table of Contents

1. Introduction 2
2. Problem Statement 2
3. Objectives 2
4. Scope of the Project 3
5. Methodology 3
6. Tools & Technologies 4
7. Expected Outcomes 4
8. Conclusion 4

1. Introduction

Modern operating systems create many logs about what is happening on the system, such as user logins, running programs, file changes, and errors. These logs are important for checking system health and detecting security problems. However, raw logs are hard to read and analyze manually.

This project will design and build a logging and reporting framework for a Linux system (Kali Linux). The system will collect, organize, store, analyze, and report important system events in a clear and useful way.

2. Problem Statement

On many systems, logs are stored in different files and formats. This makes it difficult to see what is really happening on the system. Without a central place to collect and analyze logs, security issues like unauthorized access or malware activity can go unnoticed.

This project will solve this problem by creating a centralized logging and reporting system that automatically gathers system events and produces helpful reports and alerts for administrators.

3. Objectives

The main goals of this project are to:

- Collect logs about processes, user logins, file activity, and system events
- Store logs in a structured format
- Analyze logs to find unusual or suspicious behavior
- Generate automatic reports and alerts
- Demonstrate key operating system concepts such as processes, files, and background services

4. Scope of the Project

The project will focus on Kali Linux and will include:

- Logging when processes start and stop
- Recording user login and authentication events
- Monitoring file access and changes
- Storing logs in files or a small database
- Using basic rules to detect suspicious activity
- Creating daily or weekly system activity reports
- Advanced features like a web dashboard and real-time alerts.

5. Methodology

The system will be built using Python because it is easy to use and well suited for system automation.

Development steps:

1. Study Linux log sources and system events
2. Design the system structure and data format
3. Build modules to collect logs
4. Store logs in a structured way
5. Create analysis and detection rules
6. Generate reports
7. Test the system with simulated activities

6. Tools & Technologies

- **Operating System**: Kali Linux
- **Programming Language**: Python
- **Libraries / Tools**:
  - psutil (process and user information)
  - watchdog (file system monitoring)
  - sqlite3 (log storage)
  - Django (for dashboard)

7. Expected Outcomes

By the end of this project, the system will:

- Provide a central view of system activity
- Help detect suspicious actions like brute-force logins or unauthorized file changes
- Generate clear reports for administrators
- Show practical understanding of operating systems and security monitoring

8. Conclusion

This project connects operating system concepts with cybersecurity practices. By building a logging and reporting framework, it improves system visibility and security awareness while preparing students for real-world Blue Team and SOC roles.